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calculus1_midterm_2013_solution

1.

$\lim_{x \to \infty} y = 1$

2.

$\frac{(4\sqrt{3}-1)}{6}\pi - \frac{1}{3}(2-\ln 2)$

3.

$y = \frac{1}{2}x$

4.

$= \frac{32}{2} - \frac{14}{3}\sqrt{2}$

5-(1).

$$\int_0^{\ln 2} e^2 \cos(t)dt$$

5-(2)

$\cos(\ln 2) + \sin(\ln 2) - \frac{1}{2}$

6-(1)

$2\sqrt{4-x^2}$

6-(2)

$x = -\frac{1}{2}$

7-(1).

$e^{1/ne}$

7-(2)

1

7-(3)

$\therefore a = \frac{1}{2}$

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calculus1_midterm_2014_solution

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1 - (a).

$f'(x) = 1 - \frac{2}{1+x^2} = \frac{x^2-1}{x^2+1} = \frac{(x+1)(x-1)}{x^2+1}$

1 - (b).

$1-(\pi/2)$

2 - (a).

$\therefore y = \ln\left|\frac{1+\sqrt{1+x^2}}{x}\right| = \ln\left|\frac{1}{x} + \sqrt{\frac{1+x^2}{x^2}}\right|$

2 - (b).

$\therefore\ y = -\frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2} + \ln(1+\sqrt{2})$

3. a)

$\frac{1}{2}(\ln 4)^2 = \frac{1}{2}(2\ln 2)^2 = 2(\ln 2)^2$

3.b)

1. $f(x) = \dfrac{\ln x}{x}, \triangle x = 1$

$\dfrac{1}{2}[f(1) + 2f(2) + 2f(3) + f(4)]$

2. $= \dfrac{1}{2}[0 + 2 \times \dfrac{\ln 2}{2} + 2 \times \dfrac{\ln 3}{3} + \dfrac{\ln 4}{4}]$

4-(a)

$x_{n+1} = \dfrac{x_n}{2} + \dfrac{a}{2x_n}$

4-(b)

2

5-(a)

$\dfrac{x^3}{1+x^5} \leq \dfrac{x^3}{x^5} = \dfrac{1}{x^2}$ 이므로 $\displaystyle\int_1^\infty \frac{x^3}{1+x^5}dx \leq \int_1^\infty \frac{x^3}{x^5}dx$ 이고 $\displaystyle\int_1^\infty \frac{1}{x^2}dx = \lim_{b\to\infty}\left[-\frac{1}{b}+1\right] = 1$ 로 수렴함을 이용한다.

5-(b)

$\dfrac{x^b}{1+x^a} \leq \dfrac{x^b}{x^a} = \dfrac{1}{x^{a-b}}$ 이므로 $\displaystyle\int_1^\infty \frac{x^b}{1+x^a}dx \leq \int_1^\infty \frac{1}{x^{a-b}}dx$ 이다.

$\displaystyle\int_1^\infty \frac{x^b}{1+x^a}dx$ 는 $\displaystyle\int_1^\infty \frac{1}{x^{a-b}}dx$ 가 수렴하면 마찬가지로 수렴하므로, p-series에 의해서 a-b>1 인 조건하에 수렴한다.

6.

$-\pi$

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calculus1_midterm_2015_solution

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1 - (a).

$$g'(x) = (2x - \frac{1}{2})e^{-(2x^2-x)^2} + (2x + \frac{1}{2})e^{-(-2x^2-x)^2}$$

1 - (b).

$$\therefore f'(0) = -5$$

2 - (a).

$$\therefore y = \frac{1}{2}ln\left|\frac{1+x}{1-x}\right|, \; (\text{ 단, } -1 < x < 1)$$

2 - (b).

$$L(0.01) = 0.01 \approx f(0.01)$$

3 - (a).

$$\frac{1}{2}x^2\ln x - \frac{1}{4}x^2 + c$$

3 - (b).

$$\frac{3}{8}ln3 - \frac{1}{4} = \frac{3}{4}ln\sqrt{3} - \frac{1}{4}$$

4-(a).

$$\frac{1}{3}ln(x+1) - \frac{1}{6}ln(x^2-x+1) + \frac{1}{\sqrt{3}}tan^{-1}(\frac{2x-1}{\sqrt{3}}) + C$$

4 - (b)

$$\therefore xln(x^3+1) - 3x - \frac{1}{2}ln(x^2-x+1) + \ln(x+1) + \sqrt{3}tan^{-1}\frac{2x+1}{\sqrt{3}} + C$$

5.

$$\frac{4}{3}$$

6.

4

7 - (a).

$$\frac{\tan x}{1+\tan x}$$

7 - (b).

$$\frac{\pi}{4}$$